

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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1. TITLE

Implement Modules in NodeJS

2. AIM

To implement and demonstrate built-in modules and user-defined modules in NodeJS using os, path, and custom mathematical functions.

3. OBJECTIVE

- To understand the concept of modules in NodeJS.
- To use built-in NodeJS modules such as os and path.
- To create and use a user-defined module.
- To export functions using module.exports.
- To import modules using require().
- To understand the basic use of ExpressJS for creating a server.

4. THEORY

NodeJS provides a module system that allows code to be divided into separate and reusable files. A module can contain functions, objects, or variables that can be used in another program.

NodeJS modules are mainly classified into:

- Built-in modules – Modules provided by NodeJS, such as os, path, and fs.
- User-defined modules – Modules created by the programmer for specific tasks.
- Third-party modules – Modules installed using npm, such as ExpressJS.

The require() function is used to import a module, while module.exports is used to export functions or objects from a user-defined module.

The os module provides information about the operating system, while the path module provides functions for handling file and directory paths.

The basic flow is:

Module Creation → Export → Import using require() → Function Execution

5. ALGORITHM / PROCEDURE

- Start the program.
- Import the built-in os module using require().
- Display the operating system, CPU architecture, home directory, and total memory.
- Import the built-in path module.
- Define a sample file path.
- Display the file name, directory, and file extension using path functions.
- Create a user-defined file named math.js.
- Define add(), subtract(), and multiply() functions.
- Export the functions using module.exports.
- Create app.js and import the math.js module using require().
- Call the mathematical functions and display the results.
- Install and import ExpressJS.
- Create an Express application and define a route.
- Start the server on port 3000.
- Open the server in a web browser and verify the output.
- Stop the program.

6. PROGRAM

//Built in

Os.js

```
const os = require("os");
console.log("Operating System:", os.platform());
console.log("CPU Architecture:", os.arch());
console.log("Home Directory:", os.homedir());
console.log("Total Memory:", os.totalmem());
```

path.js

```
const path = require("path");
const filePath = "/home/user/documents/file.txt";
console.log("File Name:", path.basename(filePath));
console.log("Directory:", path.dirname(filePath));
console.log("Extension:", path.extname(filePath));
```

//user defined

math.js

```
function add(a, b){
    return a + b;
}
function subtract(a, b){
    return a - b;
}
function multiply(a, b){
    return a * b;
}
```

app.js

```
module.exports = {add, subtract, multiply};
const math = require("./math");
console.log("Addition:", math.add(10, 5));
console.log("Subtraction:", math.subtract(10, 5));
console.log("Multiplication:", math.multiply(10, 5));
```

//Third party

app.js

```
const express = require("express");
const app = express();
app.get("/", (req, res) => {
  res.send("Hello from Express!");
});
app.listen(3000, () => {
  console.log("Server is running on port 3000");
});
```

7. CODE EXPLANATION

Code	Description
require("os")	Imports the built-in OS module.
os.platform()	Displays the operating system platform.
os.arch()	Displays CPU architecture.
os.homedir()	Displays the home directory.
require("path")	Imports the path module.
path.basename()	Gets the file name.
path.dirname()	Gets the directory name.
path.extname()	Gets the file extension.
module.exports	Exports functions from a module.
require("./math")	Imports the user-defined math module.

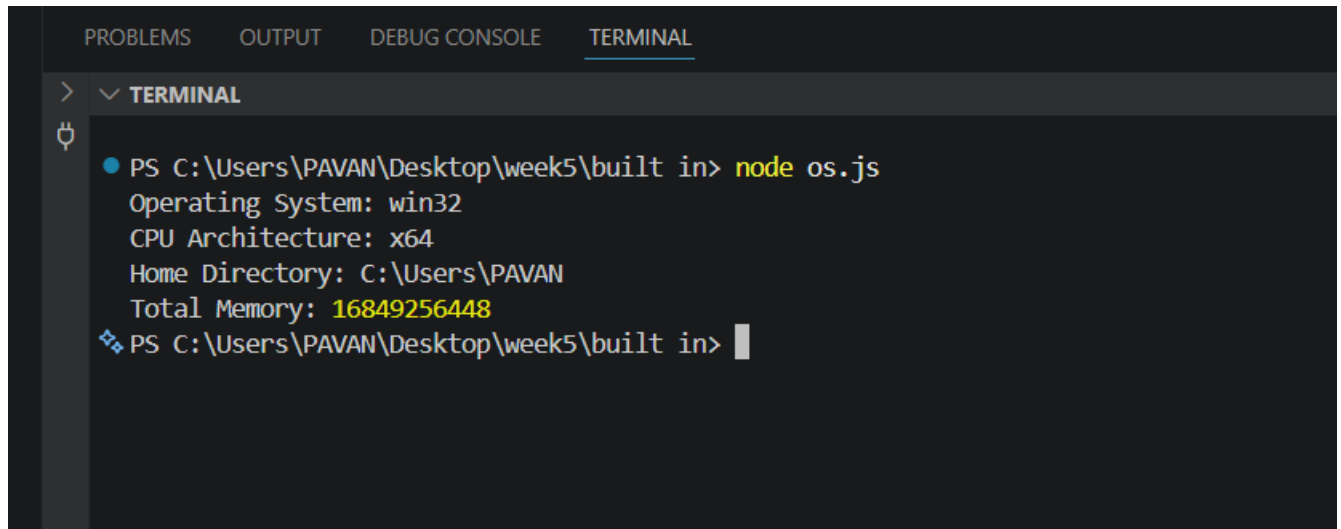
app.get()	Defines an Express GET route.
app.listen()	Starts the server on port 3000.

8. TEST CASES AND EXECUTION DATA

Test Case	Input	Expected Result	Actual Result	Status
TC-01	Run built-in module program	OS details should be displayed	OS details displayed	Pass
TC-02	Execute path.basename()	File name should be displayed	file.txt displayed	Pass
TC-03	Execute path.extname()	File extension should be displayed	.txt displayed	Pass
TC-04	Call add(10, 5)	Addition should be 15	15 displayed	Pass
TC-05	Call subtract(10, 5)	Subtraction should be 5	5 displayed	Pass
TC-06	Call multiply(10, 5)	Multiplication should be 50	50 displayed	Pass
TC-07	Start Express server	Server should run on port 3000	Server started successfully	Pass
TC-08	Open localhost:3000	"Hello from Express!" should be displayed	Message displayed	Pass

9. OUTPUT

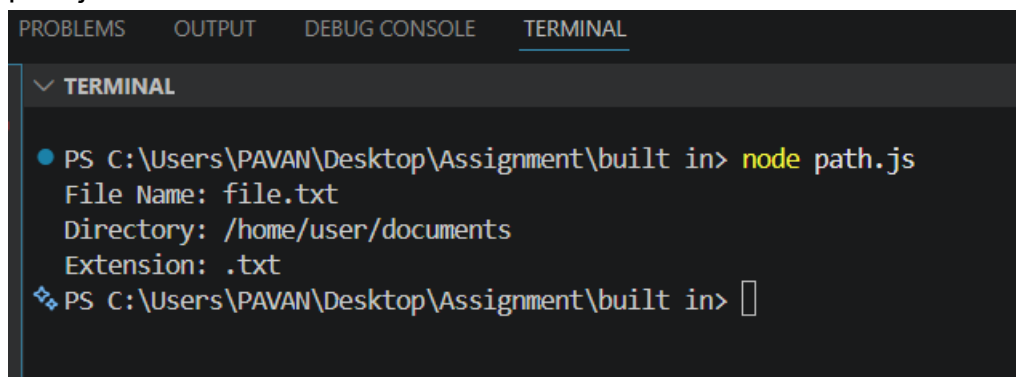
//Built in
os.js



A screenshot of the Visual Studio Code interface with the 'TERMINAL' tab selected. The terminal shows the command 'node os.js' being executed in a PowerShell prompt. The output displays system information: Operating System: win32, CPU Architecture: x64, Home Directory: C:\Users\PAVAN, and Total Memory: 16849256448. The prompt then returns to the command line.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
> ▾ TERMINAL
• PS C:\Users\PAVAN\Desktop\week5\built in> node os.js
  Operating System: win32
  CPU Architecture: x64
  Home Directory: C:\Users\PAVAN
  Total Memory: 16849256448
❖ PS C:\Users\PAVAN\Desktop\week5\built in> 
```

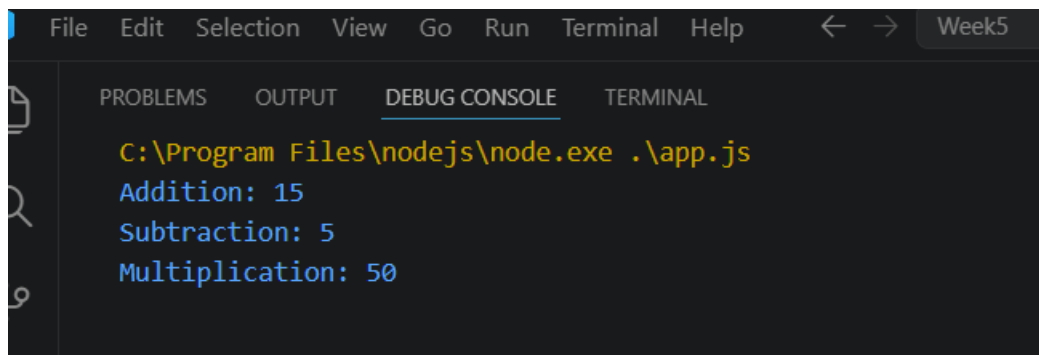
path.js



A screenshot of the Visual Studio Code interface with the 'TERMINAL' tab selected. The terminal shows the command 'node path.js' being executed in a PowerShell prompt. The output displays file path information: File Name: file.txt, Directory: /home/user/documents, and Extension: .txt. The prompt then returns to the command line.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
▾ TERMINAL
• PS C:\Users\PAVAN\Desktop\Assignment\built in> node path.js
  File Name: file.txt
  Directory: /home/user/documents
  Extension: .txt
❖ PS C:\Users\PAVAN\Desktop\Assignment\built in> 
```

//user defined

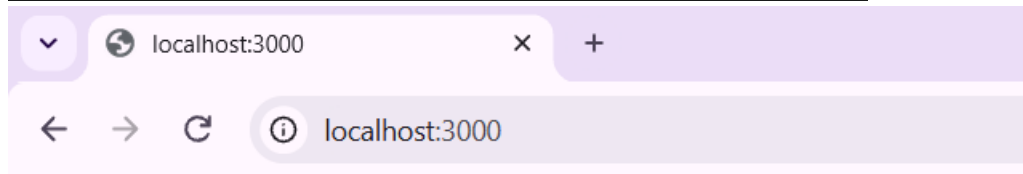


A screenshot of the Visual Studio Code interface with the 'DEBUG CONSOLE' tab selected. The terminal shows the command 'C:\Program Files\nodejs\node.exe .\app.js' being executed. The output displays the results of arithmetic operations: Addition: 15, Subtraction: 5, and Multiplication: 50.

```
File Edit Selection View Go Run Terminal Help
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
C:\Program Files\nodejs\node.exe .\app.js
Addition: 15
Subtraction: 5
Multiplication: 50
```

//third party

```
PS C:\Users\PAVAN\Desktop\fs\WEEKS\Week5\third party>  
>> node server.js  
Server is running on port 3000  
PS C:\Users\PAVAN\Desktop\fs\WEEKS\Week5\third party> |
```



Hello from Express!

10. RESULT

Thus, the modules in NodeJS were successfully implemented using built-in modules, a user-defined module, and ExpressJS. The expected outputs were obtained successfully.